

Tracing Contacts With Mobile Phones to Curb the Pandemic: Topics and Stances in People's Online Comments About the Official German Contact-Tracing App

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ABSTRACT

The Covid-19 pandemic has led to a health crisis with 90 million infections and two million deaths by the end of January 2021. To prevent an overload of medical capacities, quickly identifying potentially infected persons is vital to stop the spread of the virus. Mobile apps for tracing people's contacts seem effective, but raise public concerns, e. g., about privacy. Hence, they are contested in public discourse. We report a large-scale NLP-supported analysis of people's comments about the German contact-tracing app on news websites, social media and app stores. We identified prevalent topics, stances, and how commenting developed over time. We found privacy to be among the most debated topics discussed from various perspectives. Commenting peaked at one point in time, when public discourse centered on the potential tracing protocols and their privacy protection. We encourage further research on the link between the public discussions and actual adoption rates of the app.

CCS CONCEPTS

• **Security and privacy** → **Social aspects of security and privacy**; • **Human-centered computing** → **Empirical studies in HCI**; *User studies*.

KEYWORDS

covid-19, pandemic, contact tracing, mobile tracing app, technology appropriation, public sentiments, social media, newspaper comments

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1 INTRODUCTION

The Covid-19 pandemic is a global health crisis that has seen more than 90 million people infected and about two million dead by the end of January 2021, with new infections and deaths still on the rise [15]. Since there is no medical treatment and vaccination rollouts are just at the beginning, reducing the infection rate is paramount to prevent an overload of medical capacities.

Besides promoting passive protection such as using facemasks and social distancing, health authorities take on contact tracing as an active measure to slow down the pandemic. If a person is tested positive for the virus, their contacts are notified to self-quarantine, as they are potentially infected as well and may further transmit the virus. Such tracing must also include anonymous encounters, e.g., in supermarket queues, which is difficult to achieve with questionnaires and manual tracing. Governments worldwide therefore evaluate technical solutions such as contact-tracing apps for mobile devices. A contact-tracing app keeps track of the user's physical contacts with other people within a certain spatial range, but only of contacts using the app by themselves. If one of those traced contacts is later tested positive for the virus, the app informs its user about the risky encounter and calculates a severity score indicating whether the user should take preventive steps such as self-quarantining.

A tracing app can only trace contacts with other app users and not physical encounters with non-users. Recent research showed that about 60% of the population need to install and use the app for achieving a significant curbing effect on the pandemic [1]. Since not all citizens own a smartphone, this approach requires an even higher adoption rate among mobile phone users to work effectively. Still, many countries refrain from legally forcing the app upon their citizens, as such a move may not only face public backlash, but may also violate citizens' constitutional rights. Governments then have to rely on *voluntary* widespread app adoption, which is closely linked to the app's public visibility and perception - positively and negatively alike.

We investigate the public perception of the official German contact-tracing app called *Corona-Warn-App*, published in June 2020 [21]. The app was developed with a privacy-by-design approach from the very beginning, and has received scrutiny from the security and privacy research community [2, 23]. Instead of collecting personally-identifiable information, the app exchanges ephemeral pseudonyms only. In an attempt to boost adoption rate, the German government stressed this privacy-friendly design when promoting the app towards its citizens. Still, the app has limited

impact so far. At the beginning of January 2021, the app had been downloaded 25 million times. This number means that *at best* 30 percent of the German population installed the app on their phones [12].

In this work, we analyze the public discourse about the app over the course of five months, starting when the app was announced in March 2020, shortly before the first wave of Covid-19 infections peaked in Germany. We focus on user comments in online articles, the topics and stances commenters brought up towards the app, and how commenting activities and stances developed over time. We especially focus on how the app’s privacy-friendly design was received. Our work sheds a first light on how the public perception of the app developed and what we can take away for further app developments. As our main data source we collected more than 125,000 app-related user comments from online newspapers (tech-savvy and general audience), app stores, and social media sites.

To efficiently extract and organize people’s key arguments from this large data corpus, we used automated Natural Language Processing (NLP) with word similarity measures and hierarchical clustering. For each argument sentence the NLP retrieved its stance towards the app, i.e., whether an author argues in favor or against it, or expresses a neutral position. We also used NLP for a first clustering of retrieved arguments. After we manually curated the initial set of clusters, we coded each cluster by hand with a main topic of public discourse. By mapping out our data over time, we were able to observe how findings evolved.

We present first results showing the various topics covered by the comments with privacy being the second most prevalent topic after politics. Commenting activity reaches two peaks, one around the choice of the tracing protocols to use, and a second around the time of the app’s release. We found the times of highest engagement to differ between general news channels and news websites for tech-savvy communities.

In future research, we want to investigate the link between public perceptions of the app, as established in the news article comments, and the rather moderate adoption rate of the app. We also want to refine our NLP-supported research methodology for topic and stance analyses on large data sets from social media.

2 BACKGROUND: THE GERMAN CONTACT-TRACING APP

The *Corona-Warn-App* is the official German mobile phone application (app) for Covid-19 contact tracing built on behalf of the German government and operated by the healthcare and disease control authority RKI [11]. It is available for Android and iOS, the two most prevalent mobile operating systems. The app is open-source software with a comprehensive architecture documentation.

The app was designed to minimize the collection of personally identifiable information, along the principles of data minimization as stipulated by GDPR (EU General Data Protection Regulation, [4]). By using the DP-3T protocol, contact tracing takes place in a decentralized manner [22]. Each day the app randomly generates a fresh secret key from which it derives ephemeral identifiers. The mobile phone broadcasts these identifiers throughout the day using Bluetooth Low Energy (BLE), whereas each identifier is only sent for a short time period (about 15 minutes) before the app switches to

the next unused identifier. Because the identifiers cannot be linked to each other this technique prevents the receiving end of creating any movement profiles of the app user. Other devices with the same app in close proximity receive the identifiers and store them locally. If an encounter is later tested positive, she can voluntarily publish her secret keys used during the last 14 days to a billboard on a central server. All instances of the app regularly download these published keys and calculate all identifiers that were derived from these keys over the 14-day period. They then compare them against the identifiers the local app has received via Bluetooth (and locally stored in the app). If a match occurs, a potentially infectious encounter is identified and signalled to the app user.

The technological design of the app was heavily contested at times: actors from politics and research argued for the centralized PEPP-PT/PEPP protocol instead of the decentralized DP-3T. One main issue discussed was the different levels of privacy protection both protocols provide [2, 23].

3 DATA COLLECTION AND ANALYSIS

We collected data from three kinds of data sources:

- (1) User comments written in German on online articles about the app published by the major German news publishers *Der Spiegel*, *Die Zeit* and *Heise*,
- (2) Twitter posts relating to the app written in English or German, and
- (3) User reviews on the app in Apple’s App Store and Google’s Play Store written in English and German.

With our choice of news channels, we were able to collect data from both the general public and a technology-interested community: All three publishers have a long-standing tradition, broad coverage and high circulation of their print versions. *Der Spiegel* and *Die Zeit* focus on political reporting but also general societal topics while not consistently leaning to a particular political spectrum. By contrast, publications of *Heise* specifically address tech-savvy audiences such as IT professionals and software developers, but also people with a general interest in science and technology. Twitter was chosen as a social media channel widely used to publish news and commentary—often intertwined—to large audiences. The app stores serve to diversify our sample by contributing first-hand accounts of user experience after the app’s release.

Our collection period comprised the three month before the initial release of the app on June 16, 2020 and the two months afterwards. We automated data collection with our own crawling scripts developed on top of the Scrapy framework [6] and by using the NASTY Advanced Search Tweet Yelder [20]. We bypassed crawling restrictions by adding waiting times and reading out from web page interfaces directly instead of employing dedicated site APIs. To identify newspaper articles from which to retrieve comments, we used the search functions provided by the news outlet websites. As a pre-study, we experimented with search terms, such as *corona-virus*, *covid-19*, and *corona app*. We found, these queries retrieved a considerable portion of articles about the virus itself, partially unrelated to the tracing app. Hence, we restricted our crawler to the search query *CoronaWarnApp*. For each found news article, we collected all comments with timestamps.

For Twitter, we searched directly for pertinent posts with a similar approach. In a pre-study we again tested relevant search terms and found the terms *coronawarnapp*, *tracingapp* and *coronaapp* sufficient for our purpose. Same as for the news article comments, collected tweets were stored with their time stamps. For crawling the comments on Apple's and Google's app store, no specific search terms were required since user reviews were already tied to the tracing app itself.

The further data processing took place along the following steps, also shown in Figure 1:

- (1) **Initial cleaning** of collected data for further processing stages by removing quotes of other comments from the comment text, which are usually caused by replying. Also removal of URLs and mentioned user names.
- (2) **Separating the corpus of collected comments by language** using the language models provided by the ArgumenText API toolbox [5]. We separated German/English posts from other languages, and discarded the latter.
- (3) **Argument mining and stance detection** to extract the single arguments about the app from the corpus of comments [10] and classify the stance of each argument sentence as being either *pro*, *contra* or *unclassified* [8, 16, 24]. For this purpose, we used the ArgumenText API toolbox [5] using a BiLSTM [9] model for classification. Argument extraction took place on a per sentence base with a confidence threshold of 0.5 to decide whether a single sentence is an argument or not.
- (4) **Clustering of arguments** along topic similarities using unsupervised machine learning [3, 14] with SBERT/BERT [7, 19], which weights terms according to their relevance [13]. Clustering was done per source. We used this approach of hierarchical clustering to pair previously identified single arguments that are likely to relate to the similar topic. The similarity threshold, or threshold to cut off the dendrogram respectively [18], was chosen per source and lay between 0.45 (Heise) and 0.67 (Twitter).
- (5) **Manually curating the initial cluster set** where we first omitted clusters with arguments mainly referring to other user comments without further reasoning, clusters solely containing opinions rather than any reasoning (e.g., "it is a catastrophe", "that is blackmail"), and clusters containing statements merely repeating technical descriptions of the app. Second, we divided large clusters if topics appeared together. As a final step before manual cluster coding (see 4.1), we merged similar clusters across multiple sources that take on similar topics with different styles of reasoning.

By using an approach supported by natural-language processing techniques we were able to manage a large amount of data otherwise difficult to handle through manual efforts: Overall we collected 126, 301 comments for further processing (*Der Spiegel*: 54, 698, *Heise*: 15, 691, *Die Zeit*: 31, 775, *Twitter*: 22, 105, *Apple App Store* 534 and *Google Play Store* 1, 498). We extracted 49, 326 arguments in total as our input for the automated clustering step; 14, 888 of these arguments appeared in the output clusters generated by ArgumenText, while the remainder was omitted because of the chosen thresholds.

Conventional approaches such as coding each single comment by hand would require a significant effort for a data corpus of this size. Extracting the arguments from comments and organizing arguments with an initial clustering through automated techniques considerably speeds up the process. However, we want to point out that still some moderate effort was required to manually curate and code the 493 clusters ArgumenText automatically derived.

4 RESULTS

4.1 Main Topics of Arguments

We coded the manually curated clusters of arguments with a main topic as a further categorization. Overall, we identified the following eight main topics:

Technology Technical issues, such as the app not operating on older phone models, the accuracy of Bluetooth-based distance measurements (e.g., with respect to walls), and the consequences of politics relying on specific technical solutions. Further, comment authors cited technical difficulties as well as reports on security vulnerabilities discovered in other countries' apps, but also suggestions for technical improvements to the app.

Usability The usability of the app, especially for non-expert users or individual groups such as the elderly.

Contact Tracing Arguments reasoning about the effectiveness of contact tracing in general, regardless of the approach being used (manually by health administration workers or apps).

Government Decisions Discussions on whether the money invested into the app should rather have been spent on other projects, either generally or with respect to the pandemic.

Privacy Debates on how the app supposedly interferes with user privacy and civil rights including claims of potential abuse of the app and its data by governments or companies. The cluster comprises both criticism and positive feedback about the app.

Acceptance and Framework Arguments in this cluster debate whether the app is useful given the actual or expected adoption rates, and the voluntary model (no legal obligation to install or use the app). This cluster also contains debates on whether the app is truly voluntary, or if private entities (e.g., shops or employers) may require the app for employees or customers. Lastly, the cluster contains discussions on whether users will react upon the risk score provided by the app, i.e., seek medical attention, testing, or self-quarantine, given that it cannot be enforced in a voluntary model.

Travelling This cluster centers around data exchange between different national apps for travellers and commuters.

Politics Discussions about the existence of the pandemic, the mortality rate of Covid-19, the accuracy of PCR tests, whether governments should impose restrictions on citizens, or conspiracy theories alleging politicians to install dictatorships. This cluster also contains general criticism on capitalism (app and smartphone operating system being developed by private companies). Comments about risk and politics are usually intertwined, e.g., some argue the virus mortality would not necessitate certain political decisions.

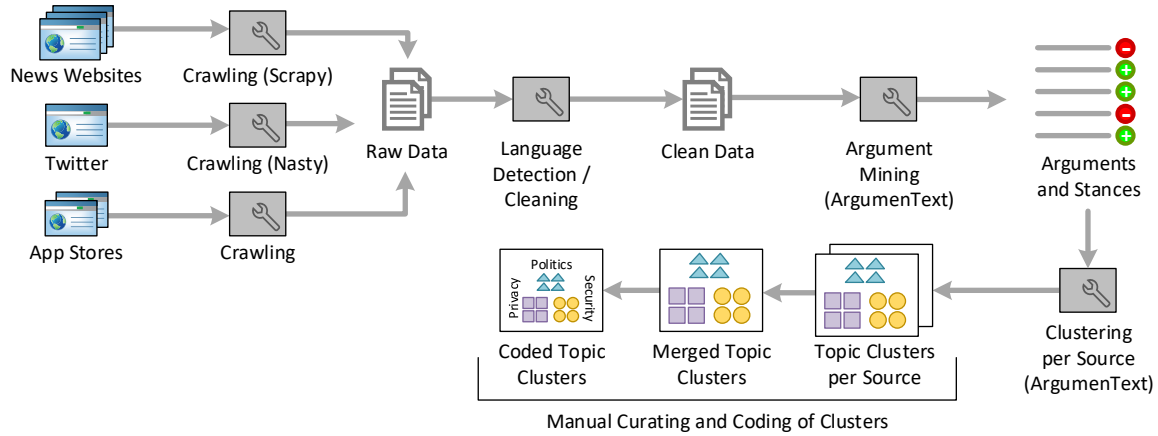


Figure 1: Illustration of the data processing. Gray boxes with wrench symbol denote tool-supported automated processing steps. In case of using a third-party tool to implement a processing step, the tool’s name is shown in brackets.

Technology	Bluetooth, works, Android, Apple, smartphone, Google, battery, GPS, error, software
Usability	clearly, intelligible, explain, fast, usage, works, easy, transparent, understand, functionality
Contact Tracing	infection chain, contact, spread, people, health authorities, help, track, test, faster, cases
Government Decisions	cost, money, millions, expensive, euros, mio, pay, country, people, development
Privacy	data, data protection, Google, Apple, privacy, people, security, Bluetooth, access, government
Acceptance and Framework	people, smartphone, install, voluntary, quarantine, infected, usage, contact, virus, faster
Travelling	smartphone, works, installed, Germany, infected, people, open, contact tracing, problem, Apple
Politics	people, virus, measures, infection, Germany, quarantine, risk, time, count, life

Table 1: Prevalent ten words in clusters categorized by main topics. Words sorted by frequency in each topic; most prevalent word first. German words translated to English.

Most arguments amount to the *Politics* topic (7,766 arguments out of 14,888 overall clustered arguments) followed by *Privacy* (2,235 clustered arguments), *Technology* (1,940 clustered arguments) and *Acceptance* (1,439 clustered arguments). We analyzed the prevalent ten words for each cluster to provide an overview of discussed aspects; stop words or terms related to the overall context of the app were removed from the list. If a word appeared multiple times with different suffixes, we considered it as a single for this evaluation, similar to [17]. The results are shown in Table 1.

The word analysis together with a deeper analysis of selected arguments indicates that a variety of aspects is covered in the eight main topic categories. To give a few examples: Technological issues are discussed in depth with respect to aspects such as how the app affects battery life or how the Bluetooth communication works. Privacy is certainly an issue to people - it is the second largest topic by number of clustered arguments - and is discussed with references to the technology (“Bluetooth”) and potential threat agents (the “government”). It is also discussed how the app can actually support identifying, testing and isolation of infected persons in practice, and the role people and their willingness to adopt the app plays for the success of the tracing approach (main topics “Acceptance” but also “Tracing”).

The relative cluster sizes were different between the various data sources, especially on the app stores. 61% of all arguments on the Apple app store were related to usability, in contrast to

0.5% on Google’s Play Store. On Google, 51% of the arguments were on technology, but only 9% on the Apple store. We further found slightly more arguments related to technology on the general-audience newspaper *Die Zeit* (13%) than on the tech-savvy *Heise* magazine (9%). The relative number of privacy-related arguments is equal (10%) for both newspapers.

4.2 Evolution of Commentary over Time

In this section, we present how the comments on the news articles developed over time. We took samples each week starting with three months before the app release, counted the number of comments, and compared the counts of positive stances to negative ones. Our topic clustering from the previous analysis was not considered; all extracted 49,326 arguments and their stances were included.

Figure 2 shows the counts per week for the news portal *Der Spiegel*, which targets the general public. The news outlet *Heise*, on the other hand, targets a tech-savvy audience with people interested in privacy and security technology. Figure 3 shows the counts for *Heise*. Both figures present positive and negative stances (green/above the baseline, and red/below the baseline, respectively). In both charts, two peaks of commenting can be noticed. One occurred around mid to end of April and a second one happened between the end of May and the beginning of June.

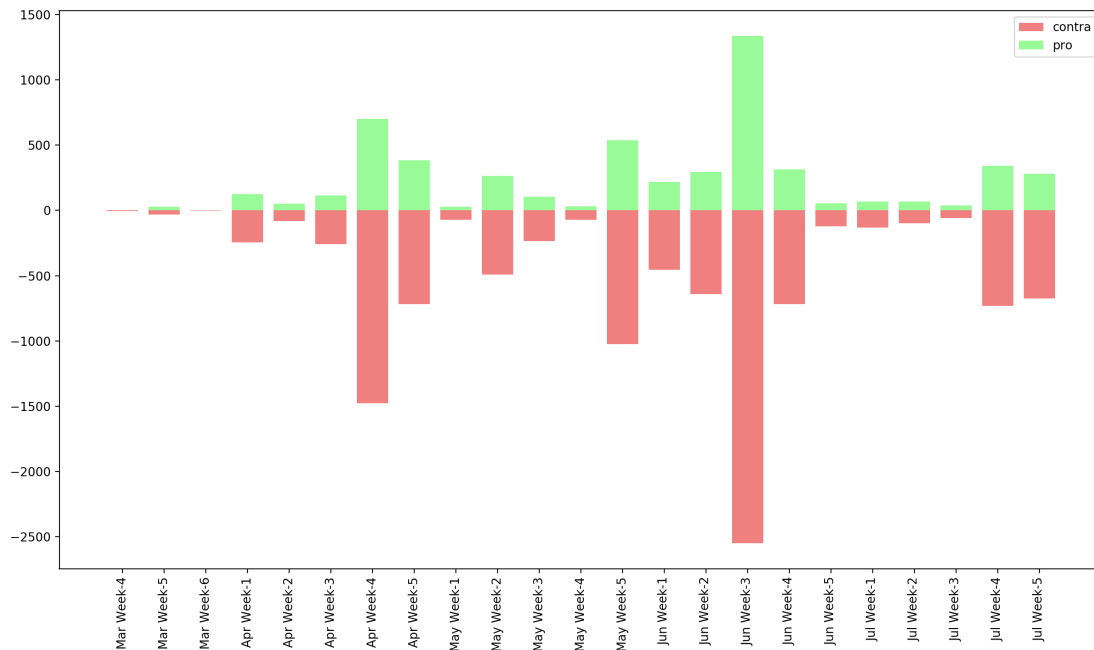


Figure 2: Number of detected stances of comments on *Der Spiegel* news articles per week. (X-axis: calendar weeks in 2020, positive Y-axis: number of positive stances, negative Y-axis: number of negative stances)

The first peak of comments in April occurred shortly after the first wave of infections reached its maximum with an incidence of 45 cases in 7 days per 100,000 population on April 5. Thereafter, the incidence declined steadily to a minimum of 3 on June 14 and flatlined at values between 5 and 10 until the beginning of a second, considerably more severe wave of infections in October 2020. Compared to other major European countries, Germany was moderately affected by the first wave: infections were contained locally, medical capacities did not collapse, and the government was not required to impose a strict lockdown with curfews. While the number of cases continued to decline in May, the government started lifting social-distancing restrictions. By June, when commenting peaked a second time in our analysis, most extensive controls had yet been revoked.

It is notable that the two commenting peaks occur earlier for the tech-savvy *Heise* news portal, in comparison to the general-purpose audience of *Spiegel*. We compared this data to the timeline of events for the app: between week 3 and 5 in April a discussion among technologists started on whether to favor the less privacy-preserving PEPP-PT protocol instead of the decentralized DP-3T protocol. At the end of April, the Germany Federal Ministry of Health opted for PEPP-PT causing considerable concerns especially among security and privacy researchers. This decision was eventually rolled back in favor of DP-3T, but caused significant discourse, as apparent from our dataset. The second peak occurs immediately before the release of the app. The source code was provided in the fifth week of May, while the app itself was released on week 3 in June. Again

it is notable that the second peak occurs earlier for the tech-savvy *Heise* news portal, i.e., close to the release of the source code. For *Der Spiegel*, the debate happened in the week of the app's release to end users.

For all observed weeks on both news channels comments contain more contra stances towards the app than supporting ones. In particular, during the end of April when the discourse around the tracking protocols and their implications on the privacy properties of the app occurred, negative stances outweigh positive stances by more than two to one for both news sites while articulating stances towards the app reaches a local maximum. We found the Apple app store to be an outlier, with 96% positive stances. On the Google store, sentiments are 62% negative, which is closer to the newspaper comments (71% negative).

5 DISCUSSION, CONCLUSION AND OUTLOOK

Our first data deliver insights into the richness of online discussions about the German corona tracing app. We found that a multitude of aspects such as the app's technology, usability or the way in which it can actually help to curb the pandemic were covered.

Letting aside the general expressions of opinions about politics, we identified privacy as the second most contested topic in our argument analysis. Moreover, we found that the controversy about which tracing protocol to use (initially preferring centralized PEPP-PT to distributed DP-3T) and the perceived implications on privacy coincides with a notable peak in online discussions and negative

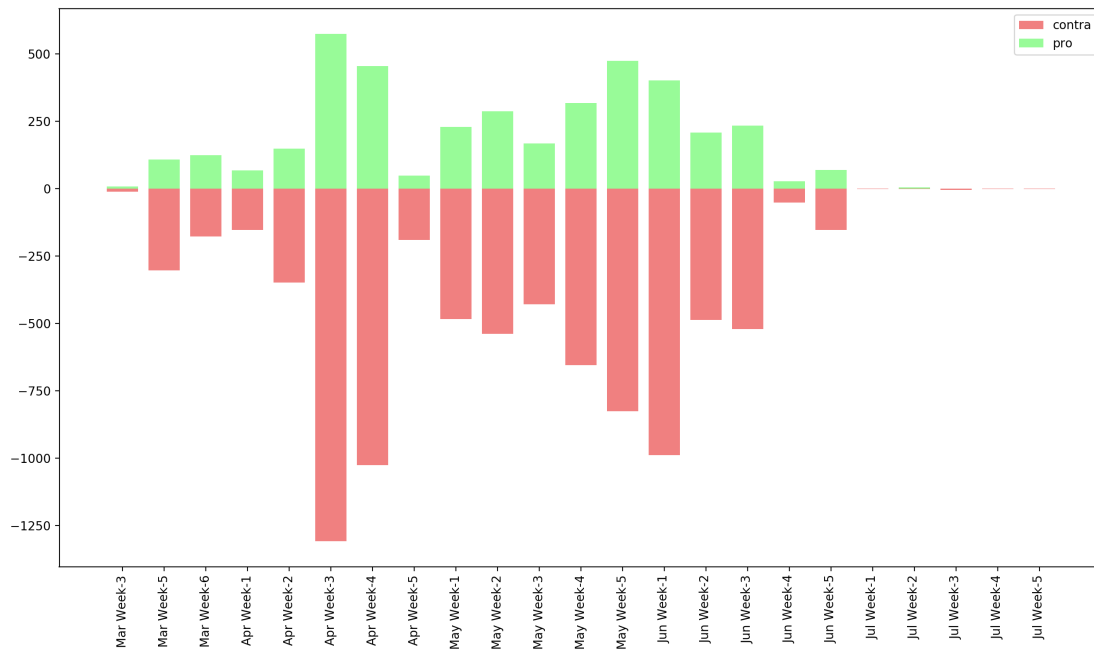


Figure 3: Number of detected stances of comments on *Heise* news articles per week. (X-axis: calendar weeks in 2020, positive Y-axis: number of positive stances, negative Y-axis: number of negative stances)

stances towards the app in both the general public but also tech-savvy communities. This finding argues for thoughtful acting when deciding on privacy techniques and communicating choices to the public. Considering that the government had to reverse its initial decision to prefer PEPP-PT to DP-3T, from a software vendor perspective's one might reason that this intense public reaction and negative public sentiments was of no avail and could have been avoided. However, the public discourse may have also had the decisive impact on the government's final decision thereby making it a cause rather than a consequence.

We observed that technology-related discussions (PEPP-PT versus DP-3T, source code release) in more general news channels like *Der Spiegel* seem to trail the tech-savvy community, but are not limited to the latter. On the other hand, reactions to practical matters such as usability or program errors in first software releases seem to be more reserved in a technophile news channel like *Heise*. Further, the stances seem to differ substantially between the communities of the two operating systems with regard to comments in the respective official app stores. We want to research these initial findings in more detail in future work as they might have considerable implications on community-specific communication strategies for technology development. Our current data set, however, is yet too limited to draw further conclusions on this issue.

As a next step, we want to refine our large-scale NLP-supported analysis with further qualitative analyses of popular news articles and related comments. We also want to look into how articles,

perhaps from different sources, are interconnected, e.g., in referring to each other.

Our work shows the power of NLP-based approaches to research a large corpus of textual data. However, we still had to engage in manual rework. For that reason, we want to advance our approach in further studies thereby delivering a methodological contribution to the HCI community.

Furthermore, we yet miss to establish a link between online discussions as researched in this work and the achieved adoption rate for the tracing app. To advance beyond this limitation of our study, we are required to contrast our research with data from user experience studies, e.g., qualitative data from focus groups with users and non-users of the app, or from quantitative surveys. So far, we can only theorize that the public discourse, in particular about the app's privacy posture, had an impact on the rather moderate adoption rate. But it is open to future research to investigate this relationship beyond our current work.

Finally, future studies may also look into how the public discourse about tracing apps was shaped in other countries and how this affected users adopting the app. While discussions about privacy are very prevalent in Germany, other topics might be considered more important in other countries.

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